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Measurable Functions

1.1 Elementary Properties of Measurable Functions

Definition 1.1.1

We endow $\overline{\mathbb{R}}$ with a topology called the order topology in the following manner. For each $a \in \mathbb{R}$ let

$$L_a = \overline{\mathbb{R}} \cap \{x : x < a\} = [-\infty, a) \quad \text{and} \quad R_a = \overline{\mathbb{R}} \cap \{x : x > a\} = (a, \infty].$$

The collection $\mathcal{S} = \{L_a : a \in \mathbb{R}\} \cup \{R_a : a \in \mathbb{R}\}$ is taken as a subbasis for this topology. A basis for the topology is given by

$$\mathcal{S} \cup \{R_a \cap L_b : a, b \in \mathbb{R}, a < b\}.$$

Observe that the topology on $\mathbb{R}$ induced by the order topology on $\overline{\mathbb{R}}$ is precisely the usual topology on $\mathbb{R}$.

Definition 1.1.2 ▶ Measurable Functions

- Suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measure spaces. A mapping $f : X \rightarrow Y$ is called **measurable with respect to $\mathcal{M}$ and $\mathcal{N}$** if $f^{-1}(E) \in \mathcal{M}$ whenever $E \in \mathcal{N}$. If there is no danger of confusion, reference to $\mathcal{M}$ and $\mathcal{N}$ will be omitted, and we will simply use the term “measurable mapping.”
- If Y is a topological space, a restriction is placed on $\mathcal{N}$. In this case it is always assumed that $\mathcal{N}$ is the σ -algebra of Borel sets $\mathcal{B}$. Thus, in this situation, a mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable if

$$f^{-1}(E) \in \mathcal{M} \text{ whenever } E \in \mathcal{B}.$$

- When Y is taken as $\overline{\mathbb{R}}$ (endowed with the order topology) and $(X, \mathcal{M})$ is a topological space with $\mathcal{M}$ the collection of Borel sets. Then f is called a **Borel measurable function**.

- When $X = \mathbb{R}^n$, $\mathcal{M}$ is the class of Lebesgue measurable sets and $Y = \overline{\mathbb{R}}$. Here, the measurability of f required that $f^{-1}(E)$ be Lebesgue measurable whenever $E \subset \overline{\mathbb{R}}$ is Borel, in which case f is called a **Lebesgue measurable function**. The definitions imply that E is a measurable set if and only if χ_E is a measurable function.

Theorem 1.1.3

If the mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable, where $\mathcal{B}$ is the σ -algebra of Borel sets. Define

$$\Sigma = \{E : E \subset Y \text{ and } f^{-1}(E) \in \mathcal{M}\}.$$

Then Σ is a σ -algebra.

Proof. Easy to verify. □

Corollary 1.1.4

A continuous mapping is a Borel measurable function.

Proof. If f is continuous, Σ contains all open sets. Since Σ is a σ -algebra, Σ contains all Borel sets. □

Definition 1.1.5

If $f : X \rightarrow \mathbb{R}$, we call the sets

$$\{f > a\} := \{x \in X : f(x) > a\}$$

the **superlevel sets of f** .

Theorem 1.1.6

Let $f : X \rightarrow \overline{\mathbb{R}}$, where $(X, \mathcal{M})$ is a measure space. The following conditions are equivalent:

- (i) f is measurable.
- (ii) $\{f > a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
- (iii) $\{f \geq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
- (iv) $\{f < a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

(v) $\{f \leq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

Theorem 1.1.7

A function $f : X \rightarrow \overline{\mathbb{R}}$ is measurable if and only if

- (i) $f^{-1}\{-\infty\} \in \mathcal{M}$ and $f^{-1}\{\infty\} \in \mathcal{M}$ and
- (ii) $f^{-1}(a, b) \in \mathcal{M}$ for all open intervals $(a, b) \subset \mathbb{R}$.